



## **EXTERNAL SOFTWARE RELEASE NOTES**

### **Release 6.0 Offline PIN Management**

**For**

**LINK Interchange Interface**

Copyright © ACI Worldwide (EMEA) Limited 2003

This document is the property of ACI Worldwide (EMEA) Ltd and the information contained herein is confidential. This document, either in whole or in part, must not be reproduced or disclosed to others or used for purposes other than that for which it has been supplied, without prior written permission or, if any part hereof is furnished by virtue of a contract with a third party, as expressly authorised under that contract.

BASE24 is a trademark of ACI Worldwide, Nebraska, USA.



## Change History

Version 1, 02/2003 (Preliminary)



## CONTENTS

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| <b>1</b> | <b>INTRODUCTION.....</b>                                 | <b>4</b>  |
| 1.1      | OVERVIEW.....                                            | 4         |
| 1.2      | SYSTEM IMPACT .....                                      | 5         |
| 1.2.1    | Device Impact .....                                      | 5         |
| 1.2.2    | External Interface Impact .....                          | 5         |
| 1.2.3    | Operational Interface Impact.....                        | 5         |
| 1.3      | SUBVOLUMES INCLUDED IN THE DELIVERY .....                | 5         |
| <b>2</b> | <b>IMPLEMENTATION CONSIDERATIONS.....</b>                | <b>6</b>  |
| 2.1      | HARDWARE DEPENDENCIES .....                              | 6         |
| 2.2      | SOFTWARE DEPENDENCIES.....                               | 6         |
| 2.3      | DEVICE REQUIREMENTS .....                                | 6         |
| 2.4      | SECURITY REQUIREMENTS .....                              | 6         |
| 2.5      | DISK FILE REQUIREMENTS .....                             | 6         |
| 2.6      | MEMORY REQUIREMENTS .....                                | 6         |
| 2.7      | PERFORMANCE CONSIDERATIONS .....                         | 6         |
| 2.8      | EXTERNAL INTERFACE IMPACT .....                          | 6         |
| 2.9      | CONVERSION REQUIREMENTS .....                            | 6         |
| 2.10     | BACKOUT STRATEGY.....                                    | 6         |
| 2.11     | OPERATIONAL INTERFACE IMPACT .....                       | 7         |
| 2.11.1   | Additional Log Messages.....                             | 7         |
| 2.12     | OPERATIONAL PROCEDURES.....                              | 7         |
| 2.13     | USER GUIDE / MANUAL UPDATES .....                        | 7         |
| 2.14     | IMPACTS ON OTHER PRODUCTS .....                          | 7         |
| 2.15     | FUTURE REQUIREMENTS.....                                 | 8         |
| <b>3</b> | <b>LOGICAL NETWORK CONFIGURATION FILE (LCONF).....</b>   | <b>9</b>  |
| 3.1      | LCONF ASSIGNS .....                                      | 9         |
| 3.2      | LCONF PARAMS.....                                        | 9         |
| 3.2.1    | EMV-PIN-UNBLOCK-SCRIPT.....                              | 9         |
| <b>4</b> | <b>CONFIGURATION REQUIREMENTS.....</b>                   | <b>10</b> |
| 4.1      | ICF .....                                                | 10        |
| <b>5</b> | <b>DATA REQUIREMENTS.....</b>                            | <b>11</b> |
| <b>6</b> | <b>INSTALLATION INTO RELEASE 6.0 ENVIRONMENT .....</b>   | <b>12</b> |
| <b>7</b> | <b>SOFTWARE RELEASE FILE LIST .....</b>                  | <b>13</b> |
| 7.1      | BASE24 VERSION 6.0 FILES .....                           | 13        |
| 7.2      | FIX FILE AND MODULE RELATIONSHIP.....                    | 13        |
|          | <b>APPENDIX A: ROLLING CODE METHODOLOGY UPDATE .....</b> | <b>14</b> |

# 1 Introduction

## 1.1 Overview

This document provides the installation instructions for the Release 6.0 Offline PIN Management enhancements for LINK Interchange Interface Support.

In order to ensure that card issuers can successfully manage the offline PIN function, e.g. ensuring that the offline PIN is synchronised with the online PIN known to the issuer, there is a requirement to implement an offline PIN Management Service. The management of the offline PIN function on the card introduces two new administrative functions for card issuers:

- Support for offline PIN change, where the cardholder wants to change their PIN to a new value;
- Support for offline PIN unblocking, where the PIN Try Counter on the card has reached zero and needs to be reset to the normal value (the PIN Try Limit).

BASE24-atm uses a twelve character left justified terminal sequence number. The LIS5 standard defines a retrieval reference number (field 037) including a six-character terminal sequence number. If the STM sequence number is more than six characters long, the rightmost six characters are now used instead of the leftmost six characters.

It is a standard that EMS event numbers should be unique within an object (with the exception of trace messages). A number of duplicate event messages exist in the LINK interchange. The BALINKS and LINKEMVS sources both raise events 2000, 2010, 2020, 2030, 2040, 2050, 2060, 2070, 2080 and 2090. The ATLINKS source raises event 5004 in 2 places. LINK raises too many critical EMS errors. These have been corrected.

ABEND numbers should be unique in a similar way to EMS numbers. ATLINKS and LINKEMVS both use ABEND number 1700. ATLINKS uses ABEND number 2100 four times. LINKEMVS uses ABEND number 2000 three times. These have been corrected.

LINK acquirers must provide full EMV data if the card details are read from the chip at the ATM. The LINK Interface has been changed to reject chip transactions without the EMV data elements.

**Note that these enhancements have been developed using the Rolling Code Methodology. Customers may need to apply fixes to their system before implementing this enhancement. See Appendix A for more details on the Rolling Code Methodology.**

## 1.2 System Impact

Consideration should also be given to the following impacts.

### 1.2.1 Device Impact

ATM devices should be EMV-capable to support PIN Change and PIN Unblock transactions.

### 1.2.2 External Interface Impact

PIN Change transactions will be identified at LINK Interchange by a processing code of '90'. PIN Unblock transactions will be identified at LINK Interchange by a processing code of '91'. Host systems that process TLF or ILF Extract files must be capable of identifying an internal BASE24 transaction code of '81' as a PIN Change transaction, and "82" as a PIN Unblock transaction. The TLF and ILF Extract records format will change if the PIN Change token (containing new PIN data) and/or the Issuer Script Results token is extracted.

### 1.2.3 Operational Interface Impact

The TLF perusal screens will display a transaction code of '81' (PIN Change transaction) and '82' (PIN Unblock transaction)

## 1.3 Subvolumes included in the delivery

The following subvolumes will be included in the release as standard:

- **SW60LINK**
  - ATLINKS
  - BALINKG
  - BALINKS
  - LINKDDLs

If EMV has been licensed then the following subvolume will also be included in the release:

- **SW60ILNK**
  - LINKEMVS

## **2 Implementation Considerations**

### **2.1 Hardware Dependencies**

Not applicable.

### **2.2 Software Dependencies**

The interface is compatible with XPNET version 3.0.

The PIN Management service will only be available in Release 6.0 systems updated with other [PIN@POS](#) changes.

### **2.3 Device Requirements**

None.

### **2.4 Security Requirements**

None.

### **2.5 Disk File Requirements**

None.

### **2.6 Memory Requirements**

Not applicable.

### **2.7 Performance Considerations**

Not applicable.

### **2.8 External Interface Impact**

Not applicable.

### **2.9 Conversion Requirements**

None.

### **2.10 Backout Strategy**

Not applicable.

## 2.11 Operational Interface Impact

### 2.11.1 Log Messages

The following event messages will have their severity changed from 3 to 2:

ATLINKS:

5001, 5003, 5007, 5023, 5025, 5027, 5029, 5030, 5034, 5032, 5031

BALINKS:

2030, 2040

LINKEMVS

2060 (changed to 2380), 2065

The following additional log messages will be required:

| Event number | Severity | Message                                                         |
|--------------|----------|-----------------------------------------------------------------|
| 0976         | 2        | FAILURE to EXPAND ATM \\\ Message - Field #                     |
| 2330         | 2        | Error while attempting to get token \\\                         |
| 2340         | 2        | Error while attempting to get token \\\                         |
| 2350         | 2        | Error while attempting to get token \\\                         |
| 2360         | 2        | Error while attempting to add token \\\                         |
| 2370         | 2        | Error while attempting to add token \\\                         |
| 2380         | 2        | Error while attempting to add token \\\                         |
| 2390         | 2        | Error while attempting to get token \\\                         |
| 2400         | 2        | FAILURE on VERIFY, Message Type \\\ MISSING FIELDS \?           |
| 2405         | 3        | Error while attempting to add token \\\                         |
| 2406         | 2        | Unable to add token \\\, no room available - Tran \\\, Card \\\ |
| 2410         | 2        | Error while attempting to get token \\\                         |
| 2415         | 3        | Error while attempting to add token \\\ - Tran \\\, Card \\\    |
| 2420         | 2        | Unable to add token \\\, no room available - Tran \\\, Card \\\ |
| 5040         | 3        | Error while attempting to get token \\\ - Tran \\\, Card \\\    |
| 5045         | 3        | Error while attempting to add token \\\ - Tran \\\, Card \\\    |
| 5050         | 2        | Error while attempting to get token \\\ - Tran \\\, Card \\\    |

## 2.12 Operational Procedures

Not applicable.

## 2.13 User Guide / Manual Updates

The BASE24 LINK (LIS5) INTERCHANGE INTERFACE MANUAL RELEASE 6.0 SW-DS969-60 has been updated.

## 2.14 Impacts on Other Products

None.



## 2.15 Future Requirements

None.



## 3 Logical Network Configuration File (LCONF)

### 3.1 LCONF Assigns

None.

### 3.2 LCONF Params

#### 3.2.1 EMV-PIN-UNBLOCK-SCRIPT

Format : Process Name:\*\*\*\*\*

PARAM: EMV-PIN-UNBLOCK-SCRIPT

Name: hhhhhhhhhhhhhhhh where h is a hexadecimal character.

Text: The script template for the PIN UNBLOCK script command, excluding the MAC. If EMV PIN authorisation is not being carried out on BASE24 any combination of hexadecimal characters can be used.

## 4 Configuration Requirements

For more details on general [PIN@POS](#) Configuration Requirements reference the External Software Release Notes Release 6.0 Offline PIN Management Authorisation Processing and Infrastructure.

### 4.1 ICF

The PIN CHANGE option in the NOT ON US and TRANSACTION ALLOWED menu (Screen 7) must be set to 'Y' to allow the ATM PIN Management transactions to be processed.

## 5 Data Requirements

If any data files will be changed as a result of bringing your source up to the fix level that this enhancement was developed against, file conversions may be necessary. These will be documented in those fixes.

For more details on general [PIN@POS](#) Data Requirements reference the External Software Release Notes Release 6.0 Offline PIN Management Authorisation Processing and Infrastructure.

## 6 Installation Into Release 6.0 Environment

The following instructions apply when implementing a Release 6.0 interface into a Release 6.0 environment. To install this release the following tasks must be performed in the order they are listed.

- 1) Analyse the fix levels of your system. If necessary, apply any fixes required or replace sources with current versions.
- 2) Restore the files listed in the software release files list (in section 7) to the target subvolumes.
- 3) Use MAKESCAN or MAKE to update the MAKE files to point to correct system, volume and subvolume
- 4) Move to the BA60MAKE subvolume. Edit CUSTMACS and set 'LINK\_ON' and 'LINK\_ATM\_ON' to \$(TRUE).
- 5) Ensure that the MAKE flags SW\_PRE\_60\_LCONF\_ON and FORTIFY\_ON are set to \$(FALSE).
- 6) If EMV is supported ensure that LINK\_EMV\_ON is set to \$(TRUE), otherwise set it to \$(FALSE).

**The remaining flags will be set according to each customer's environment and therefore should remain unaltered.**

- 7) Obey B24C to compile LINK.
- 8) Check PERUSE for successful completion of MAKE. This should result in the production of the following objects and library files:- (note: the MAKE may result in other files being built depending on what flags have been turned on).

**SW60ILNK:**

LINKEMVO

**SW60LINK:**

|         |         |          |         |      |
|---------|---------|----------|---------|------|
| ATLINKE | ATLINKO | BALINKE  | BALINKO | LINK |
| LINKCOB | LINKTAL | LINKTPLT | RQLINKO |      |

- 9) Move the LINK object into the executable environment.

## 7 Software Release File List

### 7.1 BASE24 Version 6.0 Files

If the enhancement is to be supplied as complete modules, the list below specifies the files to be included in the delivery.

|                         | Module Location | Module   |                                   |
|-------------------------|-----------------|----------|-----------------------------------|
| Standard LINK Interface | \$vol.SW60LINK  | ATLINKS  | Supplied as standard              |
|                         |                 | BALINKG  | Supplied as standard              |
|                         |                 | BALINKS  | Supplied as standard              |
|                         |                 | LINKDDL  | Supplied as standard              |
| EMV Extension           | \$vol.SW60ILNK  | LINKEMVS | Supplied if EMV has been licensed |

### 7.2 Fix File and Module Relationship

If the enhancement is to be supplied as fix files, the list below specifies the SCNs to be included in the delivery.

| Module   | SCN     |                                   |
|----------|---------|-----------------------------------|
| ATLINKS  | SW06324 | Supplied as standard              |
| BALINKG  | SW06326 | Supplied as standard              |
| BALINKS  | SW06325 | Supplied as standard              |
| LINKDDL  | SW06327 | Supplied as standard              |
| LINKEMVS | SW06328 | Supplied if EMV has been licensed |

## APPENDIX A: Rolling Code Methodology Update

*The following information first appeared in BASE24 Update (C-Vol #2, Issue #04-01, Jul-1-02).*

The Rolling Code Methodology allows enhancements and maintenance to be performed on a single version of software. Software changes are developed and unit-tested based upon all fixes completed for a module, which provides a more stable version of code resulting in higher software quality.

On a weekly basis, each fix delivered by HELP24 is reviewed by the ACI Fix Review Committee to determine the potential impact to other customers. HELP24 regularly issues Hotline letters covering critical fixes to customers who might be affected. Customers should review the HELP24 Hotline letters to determine whether their environment requires any critical fixes.

### The Challenge

The consolidation of software changes into a single version of software has increased the amount of change and dependencies between enhancements and fixes. Concurrently, customers may want to minimize software changes but without routine customer software updates, the scope and complexity of customer maintenance expands when the customer requires a specific fix.

ACI Worldwide understands that software dependencies may complicate customer projects. While routine customer maintenance is the best method to avoid issues, we realize there are practical implications with staying current. We appreciate customers' desire to receive only those software changes required to resolve a single case or implement an enhancement but that is not always possible.

ACI Worldwide recommends that customers review their current level of software changes and schedule maintenance activities regularly. Fix History User Bulletin files, containing current fix level information, are available on request through your customer manager. At any time, customers can request all current fix files, or possibly a new shipment of the source files, to allow them to get current. This will minimize version mismatch and ease implementation of future software changes.

### Improvements

We are responding to customer feedback to provide improvements for handling dependency fixes. The improvements included with 6.0:

- Separated the device dependent DDL's from the BASE files
- Updated our internal processes relating to history section standards and dependency fix tracking by including a new file for each major enhancement which will itemize all files impacted by that enhancement
- When numerous dependency fixes are detected, we will refine the list by providing only those fixes affecting products (e.g. BASE, ATM, POS, TLR, Switch.) licensed by the customer. We will continue to send all direct and prerequisite dependency fixes for all modules within the licensed product. This will ensure customers have any required prerequisite fix at their site to avoid additional requests for other software changes.

### Future Considerations

We continue to research tools to assist our customers' efforts. Efforts are under way to improve the Fix tool that will help customers with significant customizations automate their release upgrades. In addition,



the ACI Worldwide web site is being enhanced to allow customers to download mandates and fixes in a secured environment.

Any specific input on tools you believe will help would be appreciated and should be provided to your Customer Manager.